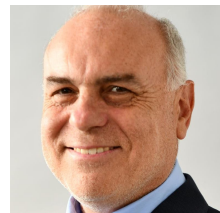


Regulations on the Migration to PQC (Post-Quantum Cryptography)

Massimiliano Sala

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Prof. Massimiliano Sala



University of Trento (Italy)

- Full professor
- **Head** of Laboratory of **Cryptography**

Italian Association **De Cifris**

- **President**

Private Consultant

- Cyber security & **Cryptography**

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Introduction

Is migration to post-quantum cryptography necessary?

Cryptography and Security

- The **security** of much of the Internet and of digital systems rests on sophisticated computational and mathematical techniques known as **cryptography**.
- The **cryptography** in common use derives its **security** and its resistance to attack from certain mathematical problems that are particularly hard for conventional computers to solve.

Quantum Risk and PQC

- A sufficiently powerful quantum computer could solve these mathematical problems, therefore putting **at risk** much of today's internet and digital systems.
- **Post-Quantum Cryptography (PQC)** refers to new **cryptographic** systems that resist **attacks** from both conventional and quantum computers.

The long journey for the PQC Migration

Migration to post-quantum cryptography (PQC) refers to the complex set of processes involved in replacing conventional cryptography with PQC. These slides offer an overview of this transition.



Current cryptography -> quantum risk-> new PQC standards -> system migrations

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Overview of the slides

These slides list the main regulations relevant to PQC migration, including recommendations and technical rules, with a focus on the US and the EU.

NIST

Technical recommendations for PQC standards and migration processes

US

Presidential executive orders and memoranda on the PQC migration process

US - CNSA

Migration plan and timeline for America National Security Services.

EU - roadmap

EU guidelines and recommendations for a coordinated PQC migration across Member States

EU - regulations

The most impactful EU regulations that explicitly address cryptography

G7 - Roadmap

Guidelines for PQC migration specifically in the financial sector

UK, France, Germany

A closer look at selected national approaches to PQC migration

Rest of the world

Regulations and timelines for PQC migration in some other countries

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NIST Recommendations

NIST - Technical body: defines the "what". Binding on federal agencies, usually followed by the private sector.

FIPS 203 · ML-KEM (2024)

Key encapsulation mechanism

FIPS 204 · ML-DSA (2024)

Digital signature

FIPS 205 · SLH-DSA (2024)

Digital signature

Hybrid approach: a possible strategy for PQC migration. It consists in pairing a classical algorithm with a post-quantum cryptographic algorithm. This ensures that security is maintained as long as at least one of the two schemes withstands attack.

	NIST	US	UE
Hybrid approach	Permitted but not mandatory.	Permitted but discouraged, with agencies free to adopt hybrid architectures based on their own risk assessment.	Encouraged as the preferred approach.
Deadline	Short-key algorithms deprecated by the end of 2030; all classical algorithms disallowed by the end of 2035.	For high-impact systems: key exchange by end of 2030, digital signatures by end of 2031. De facto deadline of end 2035 for all other systems.	High-risk use cases by end of 2030; medium-risk use cases by end of 2035.



NIST sets the algorithms and the technical deadlines; US and UE decide how far to make them mandatory.



US - Executive Orders and Memorandum

KEY MILESTONES - EO14412

2026/2027

Agencies must complete their planning and cryptographic inventory

2027

NIST pilot test

2030

High-impact systems migration: key exchange

2031

High-impact systems migration: digital signatures

2035

Migration complete



US Requirements

- Executive orders and memorandum (**NSM-10** – May 2022; **EO 14144** – Jan 2025; **EO 14306** – Jun 2025; **EO 14412** and **OMB M-26-15** – Jun 2026)
- Memorandum OMB M-26-15: full migration by 2035



The US has recently brought its timeline closer to Europe's: 2030/2031 for high-impact systems, 2035 as the horizon for completion.

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US - CNSA 2.0

COMMERCIAL NATIONAL SECURITY ALGORITHM SUITE 2.0

KEY MILESTONES

January 2027

New acquisitions into NSS must support PQC

2030

Software/firmware signing and networking equipment (VPNs, routers) must migrate to pure PQC

2033

Web/cloud, operating systems and PKI must migrate to pure PQC



NSA requirements

- Mandatory for NSS and for the vendors working with them
- Deadlines apply by product category, not by asset or system
- Oriented towards "pure" (non-hybrid) PQC. The highest parameter sets are preferred for maximum protection



CNSA 2.0 is binding only on National Security Systems.



UE - Coordinated Roadmap

KEY MILESTONES

2026

National roadmap/strategy in place;
transition begins

2030

Migration of *high-risk use cases*

2035

Migration of *medium-risk use cases*



Guidelines

- EU Recommendation 2024/1101 (EU roadmap, Apr 2024)
- Coordinated roadmap of the NIS Cooperation Group with the European Commission (Jun 2025)
- Long-term goal:
a coordinated transition across the 27 Member States



Action Plan

- Risk-based approach:
use cases divided into **high**, medium and low risk
- Hybrid schemes (PQC + conventional) permitted and encouraged
- Crypto-agility (a design principle allowing cryptography to be swapped out easily) as the guiding principle of the transition



EU: migration recommended by 2030 for high-risk use cases, with 2035 as the horizon for completing medium-risk use cases.



EU - Related Legislation

REGULATORY INSTRUMENTS

NIS2 (2022)

Requires "policies and procedures regarding the use of **cryptography** and, where appropriate, encryption"

DORA (2022)

Financial sector: "implement policies and protocols for strong authentication mechanisms, based on [...] **cryptographic** keys whereby data is encrypted based on results of approved data classification and ICT risk assessment processes"

CRA (2024)

Cybersecurity requirements for products with digital elements: "critical products with digital elements [...] and other devices for advanced security purposes, including for secure **cryptoprocessing**"

Jan 2026



COM(2026) 13: proposal to bring PQC within NIS2 "other targeted amendments to the NIS2 Directive include [...] an obligation for Member States to adopt **PQC** migration policies as part of their national strategy"

What would change



- In NIS2/DORA/CRA, the texts refer to "appropriate" or "robust" cryptography without naming PQC explicitly
- January 2026 **proposal**: an explicit obligation in national strategies (not yet adopted)



The trend appears to be towards turning the recommendation into an obligation, but this awaits the adoption of proposal COM(2026) 13.

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G7 Roadmap - Financial Sector

KEY MILESTONES

H1 2024

Quantum computing (and PQC) as a priority of G7

H2 2024

First G7 statement on planning for the risks and opportunities of quantum computing (and PQC).

January 2026

Approval and publication of the **G7 CEG Quantum Migration Roadmap**, a PQC migration roadmap for the financial sector



Roadmap

- The roadmap and its associated timelines are not intended to be prescriptive, but rather to provide information on the various activities that could be considered before the risks materialise
- Divided into **6** phases: awareness-raising, inventory, risk assessment, migration execution, testing and monitoring
- Time horizon: **2030/2032** for completing migration of **critical systems and assets**, **2035** for full completion
- Over the next 5 years, the G7 will focus on turning risk awareness into concrete action



The G7 roadmap introduces no new obligations, but rather clear guidelines to steer a timely transition

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Europe: UK, France, Germany

United Kingdom

NCSC guidelines

2028

Target date for completing an initial migration plan covering the entire organisational estate and identifying the services that depend on cryptography and will need to be upgraded to PQC

2031

Migration of highest priority services

2035

Migration of all other systems, services and products

France

ANSSI guidelines

2026

National roadmap/strategy in place

2027

ANSSI will withhold certification from products lacking PQC

2030 / 2035

Deadlines aligned with the rest of the EU.
Encourages the use of hybrid cryptography

Backs a key-exchange algorithm not selected by NIST (FrodoKEM)

Germany

BSI guidelines

2030

Migration of high-risk use cases. Hybrid approach encouraged

2031

Recommends using classical key-exchange algorithms only in combination with a PQ algorithm (for any use case)

2035

Deadline for migrating medium- and low-risk use cases (where feasible)

Backs two further key-exchange algorithms not selected by NIST (FrodoKEM / Classic McEliece)



France and Germany follow the EU line but also apply stricter domestic rules.
The UK is broadly aligned with the EU on roadmap and migration.



Outside Europe

Convergence on a 2030–2035 horizon, but with markedly different deadlines

Canada

CCCS guidelines (mandatory for government departments)

April 2026

Draw up an initial PQC migration plan at departmental level.
Annual reporting required

2031

Migration *high priority systems*

2035

Completion of migration for all remaining systems

Australia

ASD guidelines (mandatory for federal agencies)

2026

Organisations should have a well-defined PQC transition plan in place

2028

Organisations should have already begun their PQC transition, starting with **critical systems and data**

2030

Organisations should have completed their PQC transition.

ASD recommends discontinuing the use of conventional asymmetric cryptography

Asia

National strategies and guidelines

China

Expected to develop its own standards within the next three years.
Proposes combining PQC with quantum key distribution (QKD).
No clearly defined migration deadlines as yet

India

It has separate roadmaps for **critical information infrastructure** (2029 deadline) and other infrastructure (2033 deadline).
Also intends to use QKD

Singapore

Published a handbook and a Quantum Readiness Index to help companies prepare for migration.
Also intends to use QKD



Most countries worldwide are converging on roadmaps and timelines similar to those of the US and the EU. Australia is the exception, with considerably stricter guidelines. All major countries follow NIST, except China



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